

Man-made materials

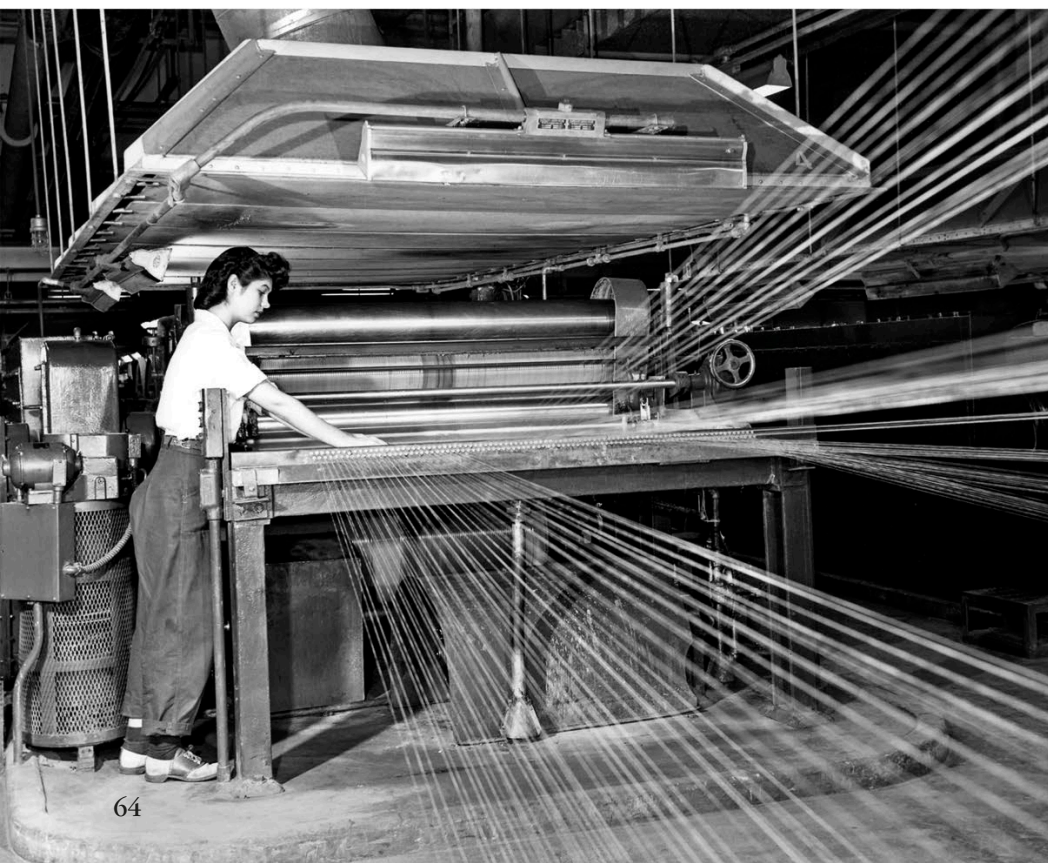
Since ancient times, we have made items—used for hunting or cooking, for example—from natural materials such as stone, clay, and wood. In the modern era, advances in chemistry and engineering have enabled us to produce man-made, or synthetic, materials, including rayon, fiberglass, and kevlar. In turn, these materials have led to inventions that benefit from a material's unique quality, from strength to elasticity.

Synthetic fibers

- **What?** Viscose rayon
- **Who?** Charles Cross, Edward Bevan, and Clayton Beadle
- **Where and when?** UK, 1892

Using their experience in soap- and paper-making, three British scientists—

Charles Cross, Edward Bevan, and Clayton Beadle—invented the viscose process. This took cellulose (an organic compound from green plants) and, through chemical treatments, turned it into a synthetic fiber that resembled silk but cost less to produce.



Kitchen tiles



Hardy floor covering

- **What?** Linoleum
- **Who?** Frederick Walton
- **Where and when?** UK, 1860s

Linoleum was invented by the British rubber manufacturer Frederick Walton as a smooth, resilient floor covering. He made it originally by coating cloth with layers of a substance containing linseed oil and other ingredients. This slowly reacted with air to form a thick, tough coating. Linoleum was plain until the 1930s, when decorative designs were added.

Making laminated, shatter-proof car windscreens



Laminated glass

- **What?** Triplex
- **Who?** Édouard Bénédictus
- **Where and when?** France, 1903

In his lab, French chemist and artist Édouard Bénédictus knocked a glass flask to the floor. It shattered but, to his surprise, the glass hung together in the same rough shape. He discovered that some cellulose nitrate (liquid plastic) in the flask had left a thin film, and this was holding the glass together. Further experiments led him to invent the world's first piece of safety glass.

Rayon production, US, 1950s